Example: Papers – Co-authors

<u>Applied as Grad Student:</u>

- G Besla : **7** co-author. [Hubble]
- E Patel: **3** co-author. [Miller, NSF]
- N Garavito-Camargo: **6** co-author[Flatiron Fellow]
- K Wagner: **7** co-author [Jansky]
- D Setton: **5** co-author[Brinson Fellow]

Co-authoring papers is ALSO important → illustrates that you can work in and leverage collaborations, and/or can support students.

Outline. (Hubble-level)

8 Pages

- Exec Summary of Past/Current and Proposed Research – 2-3 paragraphs (1/2-3/4 pg)
- Past/Current Research (3.5 - 3+1/4 Pages). [can be 2 sections]
  - Preamble: Facts that set the stage and importance of your work
  - Research Impact 1 → Section 1 + Figure 1
  - Research Impact 2 → Section 1 + Figure 2 .... Etc.
  - Summary Statement – Including Statement of Impact and Leadership
- Proposed Research (3 Pages)
  - Exec Summary of Proposal → Figure 3
  - Proposed Proj Area 1. (can have subproj/phases) → Section 1 + Figure 4
  - Proposed Proj Area 2 (can have subproj/phases) → Section 1 + Figure 5 .. Etc
  - Summary Statement – including assessment of impact and Leadership
- Feasibility/Fit (1 Page)
  - Timeline & Risk Assessment
  - Fit with Proposed Institution host → Leveraging Local Resources
  - Fit with Agency Goals (NASA/NSF etc)

Abstract Outline

Facts	
Importance of Facts	Impact within Sub Field
Problem	
Goal – identify the "key component" that will solve the problem	We propose to…
Strategy – to utilize/generate the key component , (Justify HST/JWST, explain utilizing the "target")	Suitability
Importance of Solution	Impact within Sub Field
Broader Impact	Out-of-field Impact

Exec Summary – of both current & past

Samantha Scibelli
Sagan Fellow, 2027

Facts

Importance of Facts: Your Past Research

Broad Proposal & Problem

Strategy & Feasibility

Broader Impact: NASA & Leadership

Zooming in on Prebiotic Chemistry at the Earliest Stage of Low-mass Star and Planet Formation

Planets form from disks of gas and dust around young stars whose initial physical and chemical conditions were set during the preceding evolutionary stages. Starless and dynamically evolved prestellar cores are the cold ($\sim$10 K) and dense ($\sim 10^5$ cm^{-3}) embryonic clumps of gas and dust within molecular clouds that will collapse due to gravity and external pressure to form an infant star like our Sun ($M <$ a few solar masses). My research has revealed that *prebiotic chemistry is prevalent in a large (> 65) number of starless and prestellar cores across various local (< 400 pc) star-forming environments and provides direct evidence that this molecular inventory is very likely inherited by the next stages of star and planet formation (e.g., in protostars, disks and comets).*

As a NHFP Sagan Fellow I will 'zoom in' to scales never before studied in starless and prestellar cores to answer larger questions aimed at *understanding our chemical origins*. This work leverages the unique capabilities of the 100 m Green Bank Telescope (GBT) and the Atacama Large Millimeter/submillimeter Array (ALMA) to study both high resolution kinematic structure and emission morphologies of large molecules, which will pave the way for longer-term future work with the next generation facilities (e.g., ngVLA and ALMA WSU). This research has direct ties to NASA's mission to understand how the ingredients for life develop. My extensive experience in leadership of Large Programs guarantees productivity, independence, and collaboration at local and international scales.

Exec Summary – of both current & past

Nico Garavito-Camargo
Hubble Fellow 2026

Facts

Importance of Facts

Problem

Broad Proposal

Strategy: Past Research – Impact

Strategy: Proposal – Future & Soln

Broader Impact : NASA & Leadership

Local Group Galaxies in Disequilibrium; Building New Frameworks to Constrain the Nature of Dark Matter

Context: Observations of the motions of stars and gas in Local Group (LG) galaxies, such as the Milky Way (MW) and Andromeda (M31), have revealed the existence of dark matter (DM) [1]. These observations were instrumental in the development of our current cosmological model, ΛCDM, which explains the properties of galaxies and posits that DM comprises 85% of the mass in the Universe. Yet, despite extensive searches for the DM particle, no direct evidence of its nature has been found. A promising approach is to constrain the nature of DM using astrophysical probes. Fortunately, in the LG, we have the unique opportunity to measure the properties of the DM halos of the MW, M31, and several satellite galaxies. In particular, we can measure the shape of the DM halos, which contain information on the nature of DM [2].

My Research: My research has shown that the MW's DM halo is in a *disequilibrium state*; that is, it is changing on timescales shorter than the Galaxy's dynamical time. This disequilibrium is due to the MW's ongoing minor merger with the Large Magellanic Cloud (LMC). I have proposed and demonstrated that this disequilibrium state provides a new avenue for testing the nature of DM, leveraging the wealth of multidimensional data in the LG. As an NHFP Fellow, I will investigate how DM halos respond during an ongoing halo-satellite interaction in CDM and self-interacting DM (SiDM) models. I will further develop a framework to connect the predicted halo response from CDM and SiDM models to the observed 6D kinematics of satellites and streams in the LG.

Relevance to NASA's goals: Precise measurements of the distances and velocities (6-D coordinates) of satellite galaxies, globular clusters, stellar streams, and stars orbiting in the Local Group (LG) have allowed us to measure the general properties of the DM halos of the galaxies in the LG (the MW and M31). This has been possible thanks to the exquisite measurements made by space missions such as the Hubble Space Telescope (HST) and Gaia, and ground all-sky surveys such as SDSS, DES, and DESI among others. *My research has provided new ideas and a framework to test the nature of DM in the MW using these observations and thus advances NASA's goal 'How Does the Universe Work?'.*

Previous Research Preamble

Nico Garavito-Camargo
Hubble Fellow 2026

PREAMBLE:
Facts that set the stage

Previous research: Quantification, prediction, and detection of disequilibrium during the ongoing MW-LMC interaction

The MW contains over 50 low-mass satellite galaxies and numerous stellar streams. Their orbital histories offer insights into the MW's DM halo assembly and evolution, a key observational test of the ΛCDM model. Traditionally, the MW's DM halo is modeled as spherical and static, but my research has shown that the LMC is disrupting this equilibrium. Observations across the MW—such as in stellar halo kinematics, stellar stream motions, and the warp of the MW's disk—confirm the LMC's impact. This disequilibrium affects estimates of the MW's DM distribution, total mass, and stellar orbits, while also providing an opportunity to study DM behavior during the ongoing MW-LMC merger. Below I summarize key results from my research:

Section 1 model Facts,

Highlight what YOU did

Highlight IMPACT – who else used your results?

Key Result I: The wake and reflex motion induced by the LMC. As the LMC orbits the MW, it is expected to generate a ***dynamical friction wake*** in the MW's DM halo. In [3], I predicted the properties of this wake (see upper left panel in Figure 1). This research has motivated several observational campaigns that have provided evidence of the stellar counterpart to the predicted DM wake [5–9]. I have collaborated with several observational teams [6, 9]: the left panel in Figure 1 shows the observations and predictions from my models of the overdense regions of the MW's

Past Research Impact Subsection: Section 1 model

- Header
- First Sentence(s) → The Facts
    - State the fact
    - State the importance of "the Fact" to the subfield
- Next Sentence → The Problem
    - Introduce the "Problem" → USE "However. .."
    - Why is the "Problem" is important and *critical* to solve *now*.
- Next Sentence(s): Synopsis of What you Did
    - State what you did – how did you leverage resources?
    - State what you found
    - State the importance of what you found/did to the subfield.
- Next Paragraph
    - Impact to the field of this work (Legacy Value and Deliverables)
    - How are other people engaging with this result?

Past Research Example

Ekta Patel
Hubble Fellow 2025

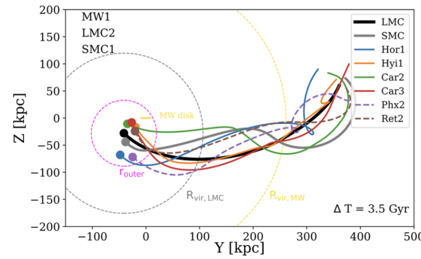

Figure 3: Orbits of all satellites identified as short-term (Ret2, Phx2; dashed lines) and long-term Magellanic satellites (Car2, Car3, Hor1, Hyi1; solid lines) using the fiducial LMC model for the last 3.5 Gyr projected in the YZ-galactocentric plane [35]. The disk of the MW lies along the z-axis. Filled circles represent the positions of all satellites today. All satellites evidence a shared orbit with the LMC (black line), providing the first evidence for a true "satellites of satellites" hierarchy.

1.4 Identifying Satellites of the Magellanic Clouds (Patel et al. 2020 – P20)

Paragraph 1:
Fact
What I did
What I found
Why it matters

Many recently discovered dwarf satellite galaxies around the MW are located in close proximity to the MCs [e.g., 1, 8, 25, 24, 23, 9, 45, 46, 47]. Using *Gaia* DR2 PMs, I determined which satellite galaxies are most likely to have a Magellanic origin by rewinding the orbital histories of all galaxies in a combined MW+LMC+SMC potential, including dynamical friction from both the MW and LMC [P20 35]. I calculated thousands of orbits iterating over measurement uncertainties in the satellite phase space properties and developed a new set of metrics to identify "satellites of a satellite" using the average distance and velocity at pericenter relative to the properties of the LMC. I concluded that approximately six satellites were accreted into the MW's halo with the MCs, as illustrated in Figure 3. This is the first confirmation of a satellites of satellites hierarchy and is complementary to ongoing projects centered on M33 satellites (see §1.2).

Paragraph 2:
Impact & Next Steps

Shortly thereafter, Sacchi et al. [38] used my classifications of LMC satellites to study the star formation histories (SFHs) of seven ultra-faint dwarfs, concluding that LMC satellites tend to show prolonged star formation (SF) and thus a later quenching time compared to non-LMC satellites, implying a connection between quenching time and environment. I am currently collaborating with N. Garavito-Camargo (Flatiron/CCA) to combine my orbital modeling techniques with his basis field expansion characterization of the MW-LMC system [15] to calculate the most accurate orbits of all MW satellites to date, including the effects of the LMC's dark matter wake, its tidal debris, and the large-scale under- and overdensities it imparts on the MW's halo (Fig. 5; Patel et al. 2023, in prep.).

Past Research Impact & Leadership Summary Statement

Ekta Patel: Hubble Fellow 2025

My past work uses the combined power of high precision astrometric data sets with high-resolution simulations of the Universe to (1) understand the influence of massive satellite galaxies on their hosts and their low mass companions; (2) look for instances of small-scale clustering via orbit reconstruction; (3) deepen our understanding of the "satellites of satellites" hierarchy; and (4) to use satellite galaxies as a tool for constraining properties of their hosts. Together these projects have revealed a modified view of the LG and its history, overturning conventional wisdom about

some of the most well-studied galaxies in the Universe, and they illustrate my ability to lead high-impact research at the interface between theory and observations using next-generation data sets.

You should know this before you start writing
→ what are the two to three impacts of your research?
→ In what ways are you leading?

Research Proposal Exec Summary:

David Setton: Hubble Fellow 2027

2.Research Proposal:

Only by combining large spectroscopic samples of post-starburst galaxies with detailed indicators of their structure, gas content, AGN activity, and bolometric luminosity can we fully understand the formation channel of the most massive galaxies in the Universe. As a Hubble Fellow, I will: **(1)** model the star formation histories of statistical samples of $z > 3$ quiescent galaxies, accounting for bolometric energy output and non-solar abundance patterns by incorporating new far-IR constraints, **(2)** use spatially resolved JWST/NIRSpec IFU and ALMA CO(2-1) spectroscopy to perform a detailed analysis of the interstellar medium in the wake of a recent starburst, and **(3)** use the revolutionary data from the Prime Focus Spectrograph Galaxy Evolution Survey to make the most precise measurements of the massive galaxy number density as a function of age, mass, and redshift to date.

Research Proposal Subsections: Section 1 model

- First Paragraph, First Sentence(s) → The Facts
  - State the fact
  - State the importance of “the Fact” to the subfield
- First Paragraph, Next Sentence → The Problem
  - Introduce the “Problem” → USE “However. ..”
  - What **opportunity** does this present?
- 2nd Paragraph: Synopsis of What you Will DO
  - State what you will do – how does this leverage resources? How does this BUILD LOGICALLY FROM YOUR Expertise
  - State what you expect to find – point to a proof of concept.
  - State the importance of what you found/did to the subfield.
- 3rd Paragraph
  - Impact to the field of this work (Legacy Value and Deliverables)
  - How do you EXPECT people will engage with this result?

Garavito-Camargo
Hubble Fellow 2026

Facts

Importance of Facts

Problem

Broad Proposal/Opportunity

What I will do

Importance of Solution

Broader Impact : NASA & Leadership

Project A: Constraining the DM halo response to satellite interactions in CDM and SiDM halos. The density profiles, kinematics, and shapes of cold dark matter (CDM) halos differ from those of self-interacting dark matter (SiDM) halos [e.g., 2]. Consequently, the response of these halos to external perturbations, such as mergers with satellite galaxies, is expected to differ [19]. Interactions between particles in SiDM generally dampen the halo's response. However, there have been no studies directly comparing the responses of CDM and SiDM halos to satellite interactions. Such a study would allow us to compare the observed properties of the MW's halo response to the passage of the LMC, presenting a unique opportunity to constrain the properties of DM in the MW, with broader implications for the fundamental nature of DM.

I will utilize Basis Function Expansions[2] to characterize the DM halo's response to satellite interactions, similar to my work in [4]. BFEs have been extensively used to understand the dynamics of DM halos in disequilibrium [20]. Figure 4 shows an example of a DM halo perturbed by an LMC-like satellite (note that the particle data from the satellite was removed to isolate the MW-like DM halo response).

The halo response is primarily captured by a dipole, induced by the satellite near pericenter passages (as shown in Figure 1), affecting the density and kinematics of the halo, along with the dipole response (see Figure 4). I will make specific predictions on the relative differences in the dipole and quadrupole amplitudes in the density, potential, and acceleration fields in CDM and SiDM. These results will provide the first quantification of the MW-LMC response in SiDM and its deviations from CDM, highlighting how BFEs can be effectively used to compare DM halo responses across different DM models.

Deliverables and Legacy: I will publish these results in Fall 2025. I have already developed all necessary methodologies in [4] and through ongoing projects (see Current Research). I will also release code and tutorials to use BFE in cosmological simulations.

Figures

You want a key figure that summarizes the proposal

\# Figures
DS: 5
NGC: 7
SS: 4
KW: 4
EP: 5

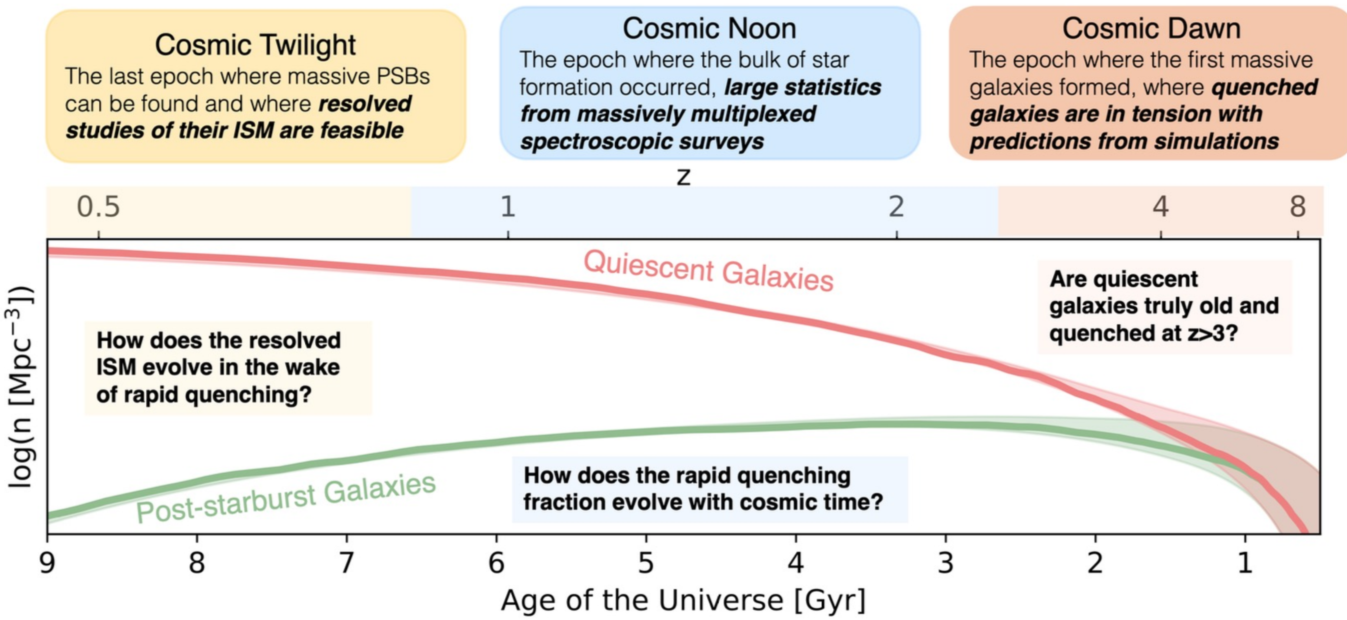

Figure 3: The epoch of quenching when post-starburst galaxies (green) were significantly populating the quiescent sequence (red) took place many Gyr in the past. As a Hubble Fellow, I will study quiescent galaxies across cosmic time in these three key epochs. **I will place stringent constraints on their star formation histories at cosmic dawn, the physical processes that regular quenching at cosmic twilight, and the connection betweeen young and old populations at cosmic noon.**

Research Proposal Summary Statement

David Setton : Hubble Fellow 2027

Together, these projects will result in a novel benchmark to which models of quiescent galaxy formation can be compared. By measuring the star formation histories across cosmic time, informed by the insights from the panchromatic study of highly resolved systems, I will link these populations of rapidly quenching galaxies to their progenitors and descendants. These studies—with data that is either in hand or set to be observed in the coming year—will enable a greater understanding how the dense cores of the most massive elliptical galaxies formed, and how the physical processes that regulate star formation and quenching evolve over cosmic time.

Risk Assessment

Kevin Wagner Sagan Fellow 2020

Scientific Yield from Planet Discoveries and Non-Detections: The planets discovered in this survey will occupy a new parameter space for imaged exoplanets (Figures 1, 3). As no sub-Jovian exoplanets have been imaged, this would open up a new class of planets for direct atmospheric studies. Furthermore, the existence of giant planets within the habitable zones of our nearest stars would dramatically influence future searches for habitable planets (or moons) within these systems. *On the other hand, the non-detection of giant planets would be consistent with the exciting possibility of smaller planets occupying the habitable zones of our nearest stars.*

Risk Assessment: The primary risk to this project is associated with the instrumental component; however, this risk is mitigated by the simplicity of the task. Without being used for coherent beam combination, the pupil-mirrors typically sit idle, and programming a ~10 Hz chopping pattern will be straightforward compared to their typical kHz active control loop. On the observational side, weather-loss will not be a significant issue with queue-mode observing. Uncertainties in the occurrence rates might be viewed as a risk; however, the actual number of detected planets will test the higher occurrence rate extrapolations with ≥95% confidence.

Feasibility

Samantha Scibelli
Sagan Fellow, 2027

Proposed Research Products are clearly articulated

RISK ASSESSMENT

3 Program Timeline

Now	• Continue observations for the Large Program GLUCOSE and **publish** first results from the initial discovery paper (**Scibelli et al.,** *in prep*) • Anticipate the arrival of the remaining ALMA Cycle 12 data
Year 1 (2026/27)	• Continue ALMA Band 1 reduction and imaging • **Publish** the ALMA maps of COM distribution around L1544 • The GLUCOSE line survey is set to officially 'end' in the spring of 2027; Co-Is and/or students will lead their own studies (e.g, isotope fractionation and isomer comparisons), under my supervision. • Propose for GBT follow-up observations to expand frequency coverage
Year 2 (2027/28)	•Analyze and **publish** ALMA Band 1 observations (and JWST comparison) of dynamic environment around prestellar core IRAS 16293E • Write and **publish** GLUCOSE overview and line inventory paper. • Continue to write follow-up proposals based on new discoveries from my large datasets (in GLUCOSE, my large Perseus survey, or ALMA)
Year 3 (2028/29)	• **Publish** follow-up study(studies) on new detections/mapping results • I will propose for an ALMA Large Program to observe COMs in a larger number of cores, setting me up for a competitive faculty position

Risk Assessment: The scientific implications of my work will have a high impact in the star formation and astrochemistry communities, and yet is *very low risk* because the data needed for each proposed project is already delivered or in progress.

Feasibility

Kevin Wagner
Sagan Fellow, 2020

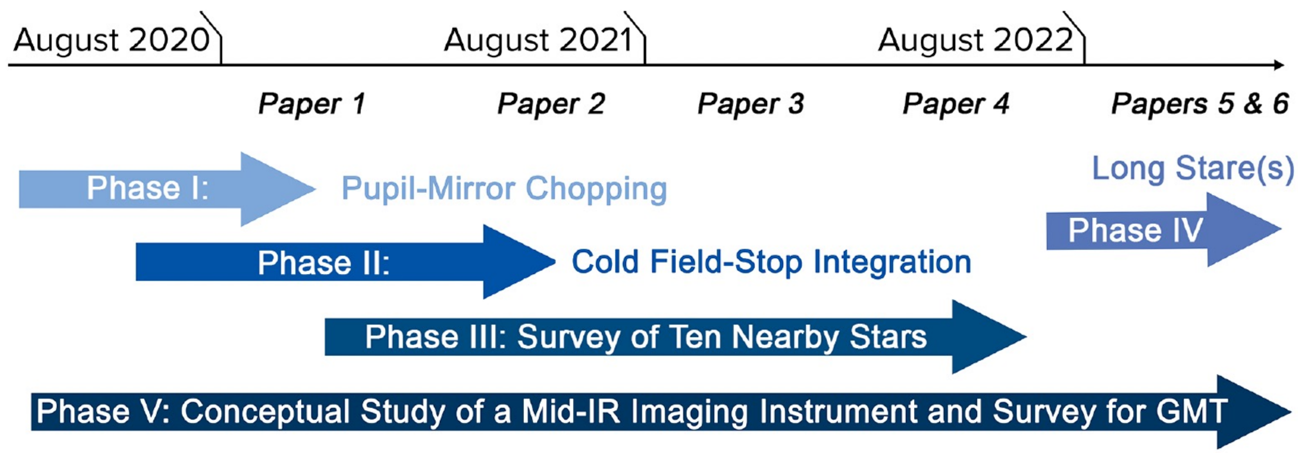

Figure 4. Timeline of proposed workflow to upgrade the mid-infrared exoplanet imaging capabilities of the LBT (Phases I and II), to conduct both a deep survey of ten nearby stars (Phase III) and a very deep exposure on a select target (Phase IV), and to complete a conceptual study for extending this work in the future with the upcoming GMT (Phase V).